



Data Analysis Portfolio

Customer Segmentation & Lookalike Audience Strategy

A User Growth Analytics Case Based on public ecommerce behavioral data —

Identifying high-value users through K-means clustering and TGI analysis, and developing actionable lookalike audience activation strategies.

Iris Xia

iris.daxia@gmail.com

Project Overview



Business Objective

Identify users with higher category expansion potential to increase customer lifetime value (CLTV) and category penetration.

Data Source

Taobao (Chinese e-commerce platform) User Behavior Dataset

Tech Stack

BigQuery

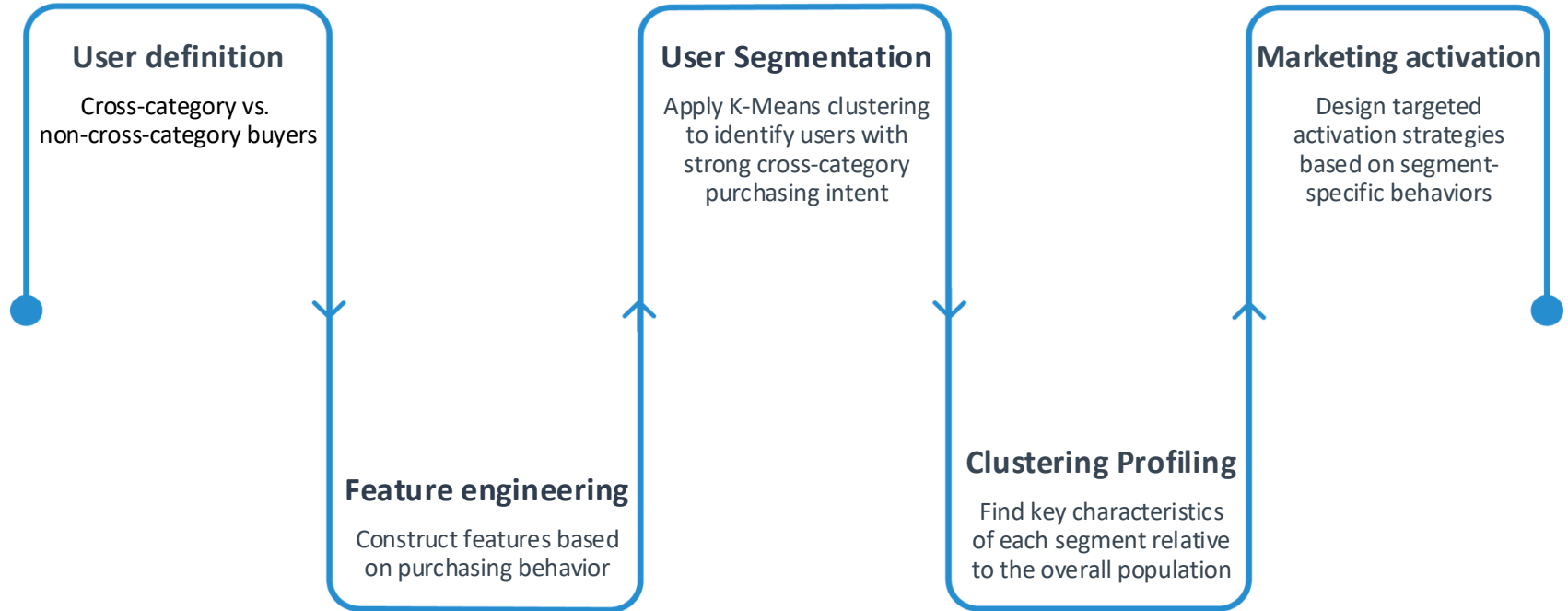
SQL

Python

K-Means

Analytical Framework

A five-stage analytical framework connecting raw behavioral data to actionable marketing strategies.



Each stage builds upon the previous one, creating a complete and interpretable path from data to business action.

Feature Engineering: 5 Behavioral Dimensions

Constructed 10 features to capture users' purchasing and exploratory patterns from multiple perspectives



Recency

Days since last purchase



Purchase frequency

Longest active streak · purchase days · average events per active day



Exploration behavior

Distinct products viewed · distinct categories viewed · product view depth



Purchase intent

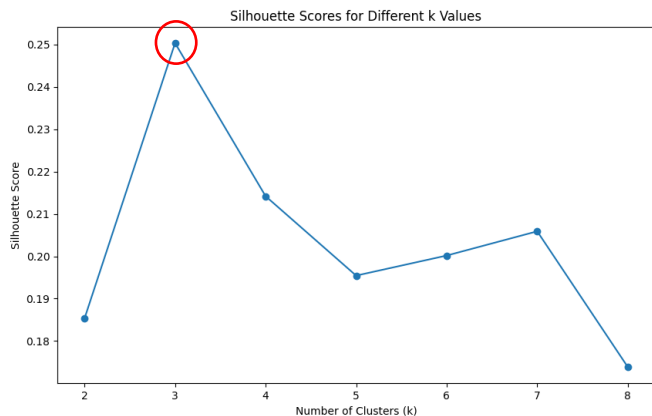
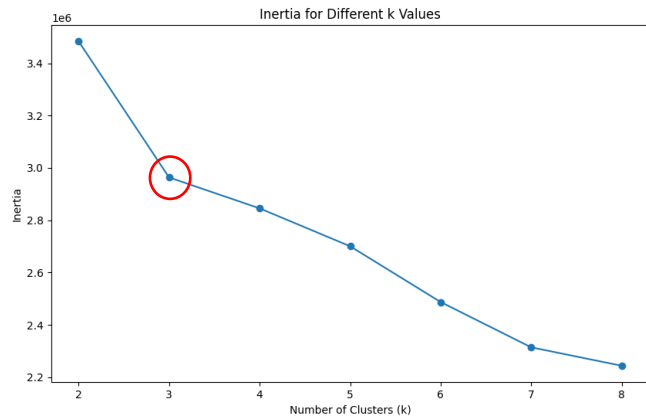
cart-to-view ratio · favorite-to-view ratio



Initial Category Breadth

number of categories in first purchase

User Segmentation: K-Means Model Evaluation



Key findings

Combining the Elbow Method and Silhouette Score analysis, **K = 3** was selected as the optimal number of clusters.

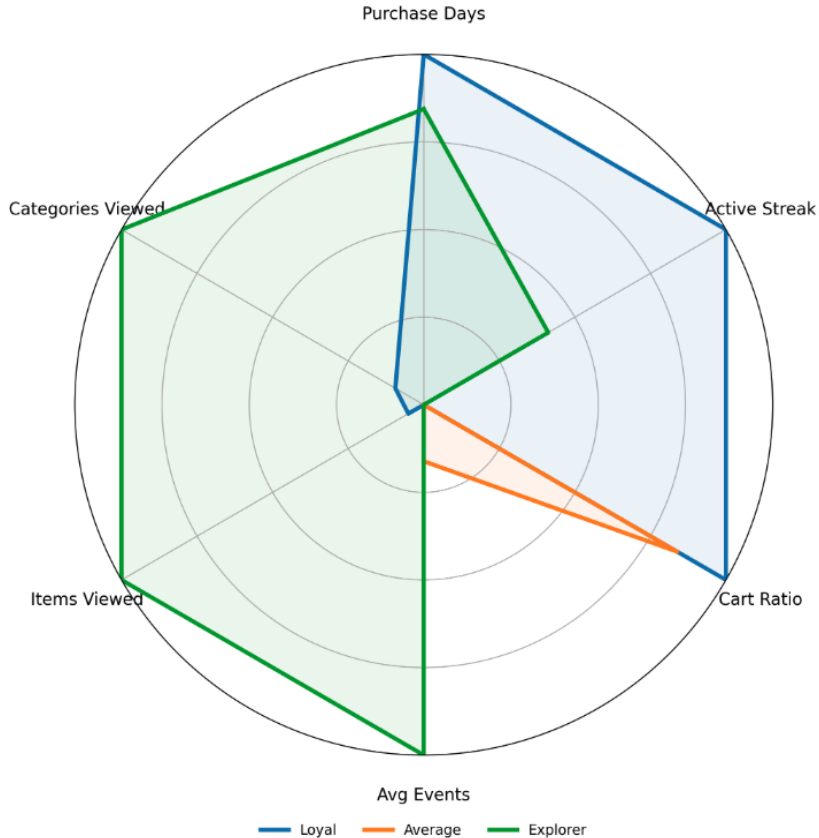
Elbow method

Model error decreases as the number of clusters increases. A clear elbow point appears at **K = 3**, where additional clusters provide diminishing returns.

Silhouette score

The highest silhouette score (**0.25**) was achieved at **K = 3**, indicating strong within-cluster similarity and clear separation between clusters, which is the best overall clustering performance.

User Segmentation: Results



Three user clusters

Based on the behavioral feature matrix, K-Means identified 3 user segments with significantly different behavioral patterns.

Loyal

Frequent purchases, more active, higher add-to-cart engagement

Average

Slightly higher add-to-cart engagement

Explorer

High cross-category browsing activity, lower purchase intent

Clustering Insights: Two Actionable Segments

Compared with the overall user base, **Loyal** and **Explorer** users exhibit different behavioral patterns and business value, making them the most actionable segments.

The **Average** segment shows limited differentiation and is therefore expected to generate lower incremental returns from targeted marketing efforts.

Loyal

High repurchase · Strong loyalty · High spending potential

Explorer

Broad browsing · Multi-category interest · Conversion opportunity

Cluster-Level TGI Insights

LOYAL

Purchase Days	139
Cart Ratio	117
Active Streak	108

Frequent purchasers with strong purchase intent and retention.

AVERAGE

Cart Ratio	110
Categories First Purchase	100
Avg Events	99

No strongly differentiated behavioral patterns.

EXPLORER

Items Viewed	283
Categories Viewed	234
Avg Events	231

Highly engaged browsers with strong exploration behavior.



Segment-Specific Marketing Activation Strategies



Loyal Activation

- Premium product bundle recommendations
- Membership tier upgrades and exclusive benefits
 - Cross-selling of high-value products



Explorer Activation

- Personalized new-product recommendations
 - Cross-category coupon incentives
 - Discovery-page content campaigns

LOOKALIKE AUDIENCE

FIND NEW USERS
WITH SIMILAR BEHAVIORS



TARGET USERS
Cross-Category
Buyers



**LOOKALIKE
AUDIENCE**
High Potential
Users



**BEHAVIOR
ANALYSIS**



**PATTERN
MATCHING**



**AUDIENCE
EXPANSION**

Lookalike Audience Expansion

Strategy Overview

Using the identified Loyal and Explorer segments as **seed audiences**, users with highly similar behavioral characteristics were identified among non-cross-category buyers to expand the pool of potential high-value customers.

14,486 users

Explorer lookalike audience

Potential cross-category buyers exhibiting strong exploration behavior.

3,081 users

Loyal lookalike audience

Potential cross-category buyers exhibiting strong loyalty characteristics.

Key Takeaways

1 End-to-End Analytics Workflow

Built a complete analytical pipeline from raw behavioral data to business recommendations using BigQuery and SQL.

2 Meaningful User Segmentation

Identified 2 highly differentiated cross-category buyer segments: loyal and explorer users

3 Actionable Lookalike Audience Selection

Developed a lookalike audience pool covering **17,567 users**, providing a scalable target group for future marketing campaigns.

4 Business Impact Orientation

Translated analytical insights into actionable activation strategies, demonstrating the practical value of data-driven decision-making.

Future Opportunities: Validate the proposed activation strategies through **A/B testing** and leverage predictive modeling to identify users most likely to make cross-category purchases.

